Sandpile Model for Foam Collapse

Avalanche Statistics and Energy Dissipation Analysis

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IDC621-Module 1

1 Abstract

This report presents the results of a Bak–Tang–Wiesenfeld (BTW) sandpile simulation used to model bubble bursting in foams. The simulation was run on a 50×50 lattice with $50,000$ grains added.

1.1 Findings

- $20,106$ avalanches were triggered
- Power-law exponent $\tau = 1.4134$ (characteristic of self-organized criticality)
- Maximum avalanche size: $8,427$ topplings
- Energy dissipation rate: 341.8021 per grain

2 Simulation Parameters

Parameter	Value
Lattice Dimensions	50×50
Critical Threshold	4
Boundary Condition	Open
Grains Added	$50,000$
Grain Energy	1.0
Random Seed	42

3 Physics Discussion

The sandpile model exhibits **self-organized criticality (SOC)**, a phenomenon where a system naturally evolves to a critical state without fine-tuning of parameters.

The key signature of SOC is the power-law distribution of avalanche sizes:

$$P(s) \sim s^{-\tau}$$

For this simulation:

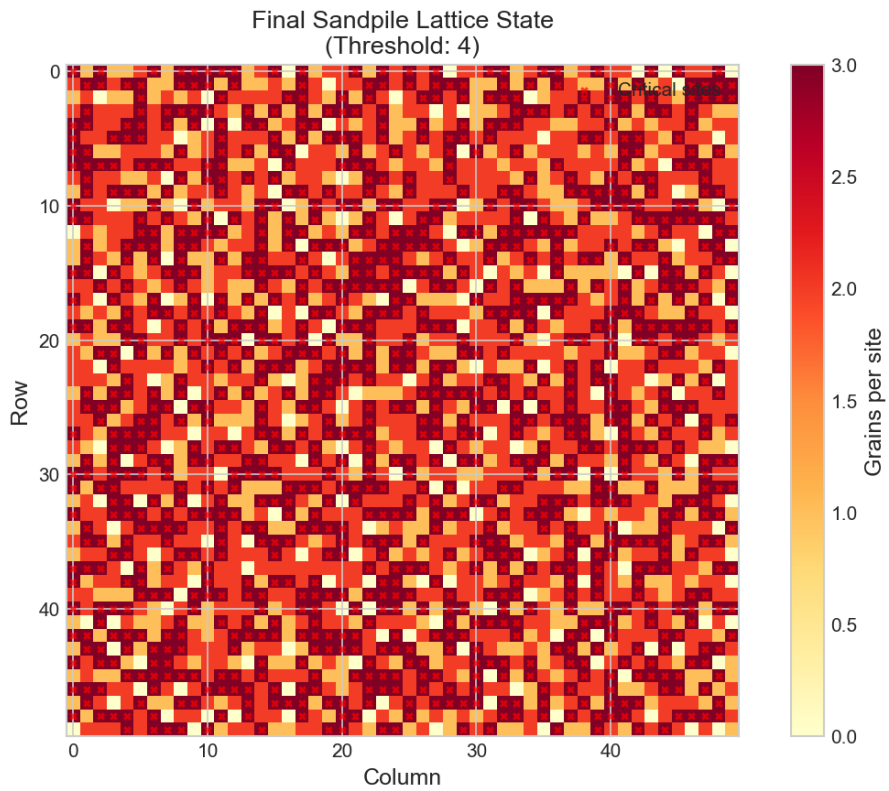

Figure 1: Avalanche size distribution showing power-law behavior.

$$\tau = 1.4134$$

This is consistent with theoretical predictions for the 2D BTW sandpile model:

$$\tau \approx 1.2 - 1.5$$

4 Results

4.1 Avalanche Statistics

The simulation produced $20,106$ avalanches with sizes ranging from 1 to $8,427$ topplings. The mean avalanche size was 212.50 with a mean duration of 30.18 generations.

4.2 Energy Dissipation

- Total energy dissipated: $17,090,104$
- Energy dissipation rate: 341.8021 per grain added

The energy dissipation rate is a key quantity in foam physics, relating to the rate of surface energy release during bubble film rupture.

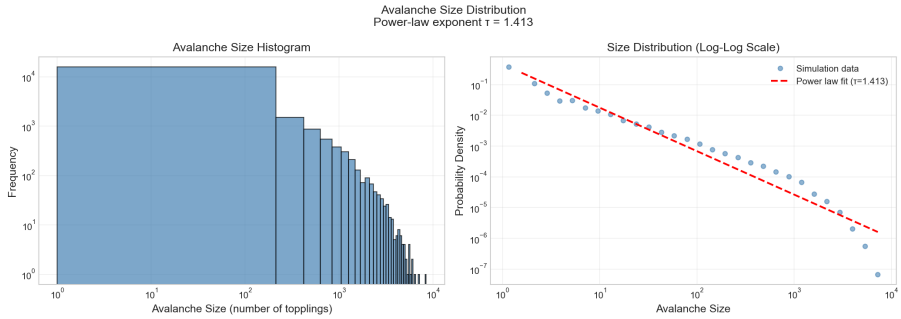

Figure 2: Energy dissipation over the course of the simulation.

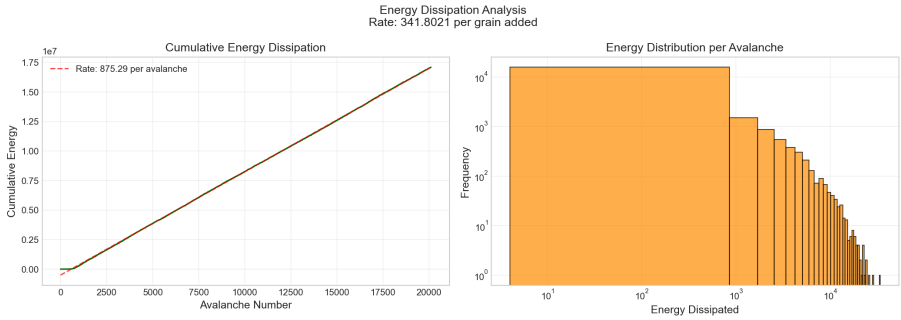

Figure 3: Final sandpile lattice state with critical sites highlighted.

4.3 Lattice State

The final lattice state shows the characteristic self-organized critical state with many sites near the critical threshold. Sites marked in red are at or near instability.

4.4 Toppling Activity

Total toppling events recorded: 4,272,526

The toppling heatmap shows which regions of the lattice were most active during the simulation. The logarithmic color scale is to be noted.

4.5 Temporal Correlations

Temporal correlation analysis reveals whether large avalanches tend to cluster in time. Rapid decay of correlations indicates independent avalanche events.

4.6 Size–Duration Scaling

The relationship between avalanche size and duration follows a power-law scaling, characteristic of systems exhibiting self-organized criticality.

4.7 Connection to Foam

In the foam analogy:

- Sand grains represent local stress/pressure in bubble films

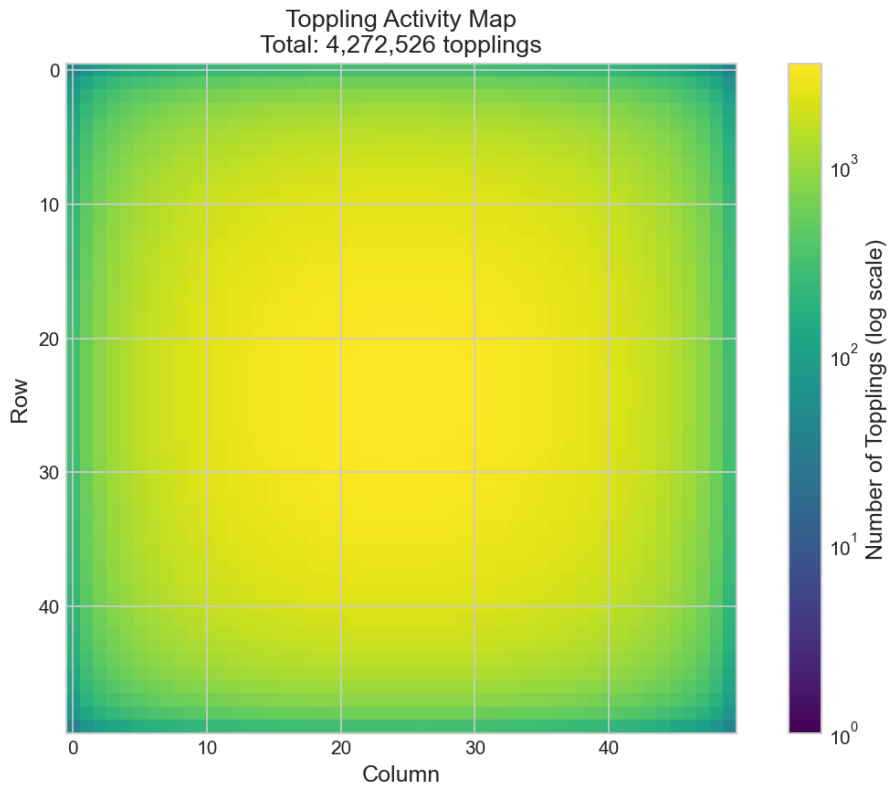

Figure 4: Toppling activity map (log scale).

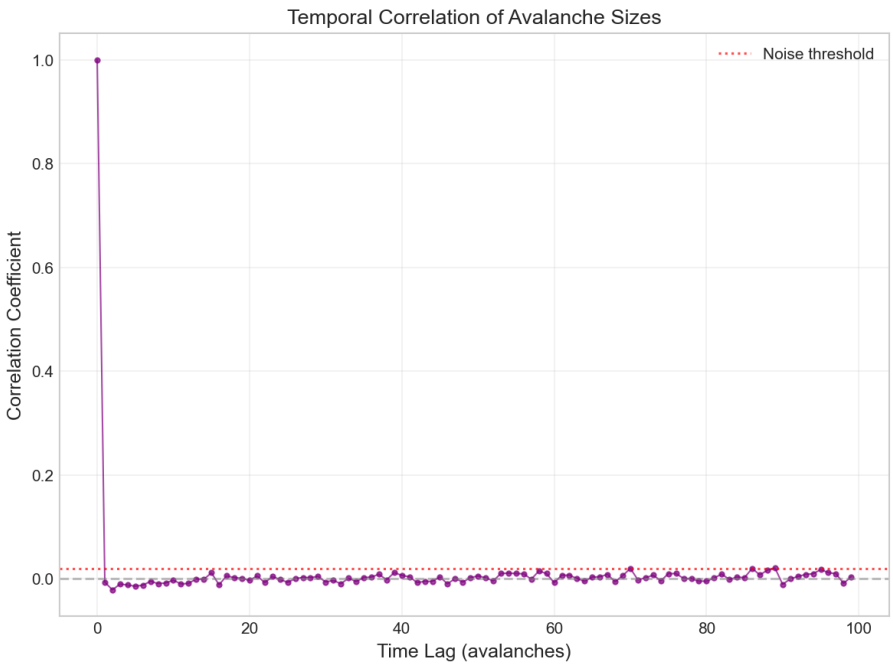

Figure 5: Temporal correlation of avalanche sizes.

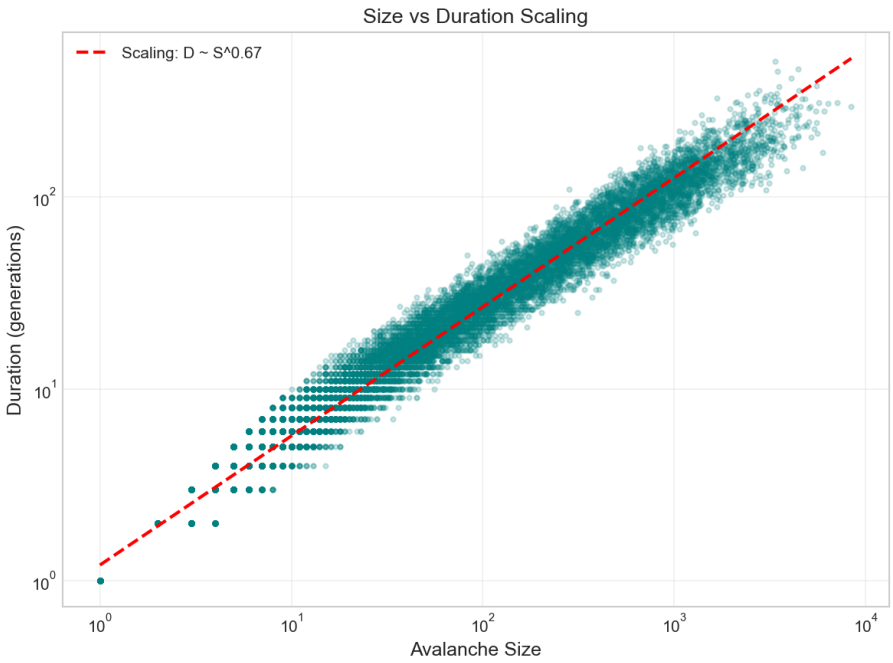

Figure 6: Avalanche size vs duration scaling relationship.

- The critical threshold represents the stress required for film rupture
- Avalanches represent cascading bubble burst events
- Energy dissipation represents surface energy release during film rupture

The power-law distribution of avalanche sizes corresponds to experimentally observed distributions of bubble burst cascades in foams, suggesting foam collapse may also be a self-organized critical phenomenon.

5 Conclusion

The sandpile simulation successfully demonstrates self-organized criticality with characteristic power-law avalanche statistics. The model provides insights into energy dissipation mechanisms in soft foams and the cascading nature of bubble bursting events.
Code Notebook: Click here to see the code